Power Plant Engineering: Energy Balances, Cycles, and Heat Transfer

MECHANICAL-ENGINEERING · power-plant-engineering · BoardPrep lesson v1

Official examination scope

The August 2026 PRC MELE program assigns Power Plant Engineering its own subject and relative weight. This lesson teaches selected energy and heat-transfer reasoning; actual plant operation, safety, emissions, and design require current applicable rules and qualified engineering review.

Learning objectives

- Apply an energy balance with a clearly defined system boundary.
- Calculate heat transfer through series thermal resistances.
- Compare cycle performance without confusing ideal and actual plant behavior.

1. System boundaries and energy balances

Choose the system boundary before writing an energy balance. For a steady-flow device, account for heat transfer, shaft work, mass flow, and changes in enthalpy, kinetic energy, and potential energy as relevant. State whether a term is neglected and why. A boiler, turbine, condenser, pump, and feedwater path have different energy interactions even though they belong to one plant. Keep mass and energy balances distinct, preserve consistent units, and compare the computed result with the expected direction of heat or work transfer.

2. Heat transfer and thermal resistance

For a steady one-dimensional resistance model, heat rate equals temperature difference divided by total thermal resistance. Series resistances add when the same heat rate passes through each layer and contact conditions fit the model. Parallel paths combine differently. The formula is an engineering analogy to conduction and does not by itself describe transient storage, radiation, convection correlations, or two-dimensional heat flow. Verify units: kelvins divided by kelvins per watt gives watts. Check whether surface and fluid-film resistances are included in the stated data.

3. Cycles, efficiency, and plant performance

A thermal cycle converts supplied heat into work while rejecting some energy to a lower-temperature sink. Thermal efficiency is net useful work divided by heat input for the stated boundary and cycle. Ideal cycle relations help compare limits; actual performance also reflects component inefficiency, pressure drop, heat loss, off-design operation, and auxiliary loads. Compare efficiencies only when system boundaries and input definitions match. A plant heat rate or load curve may include operating constraints that a simple thermodynamic cycle omits.

4. Fluid machinery, HVAC, and hydroelectric systems

Pumps and turbines exchange shaft work with a fluid; their analysis relates flow, head, pressure, elevation, and losses. Fans and compressors also require a state model appropriate to the fluid and process. Refrigeration transfers heat from a colder region to a warmer one by supplying work; psychrometric loads depend on both temperature and moisture. Hydroelectric power draws on water flow and head, with conversion losses and environmental and operational constraints. Actual equipment selection requires system curves, operating ranges, manufacturer data, and applicable rules.

Calculate heat rate across two resistances

Question: Two thermal resistances of 0.20 K/W and 0.30 K/W are in series across a 50 K temperature difference. Find the heat rate in the stated steady model.
1. Total resistance = 0.20 + 0.30 = 0.50 K/W.
2. Heat rate = T/R = 50 K ÷ 0.50 K/W.
3. Check that units reduce to watts.
Answer: 100 W.
Check: This assumes the supplied resistances are the complete series path and the process is steady.

Common misconceptions

- Cycle efficiency can be interpreted without defining heat input and work output.
- Series and parallel thermal paths use the same resistance rule.
- An ideal-cycle result predicts actual plant operation exactly.

Practice patterns

- Draw the control volume and sign convention.
- Write every resistance and its unit before combining.
- Name losses and operating conditions omitted from an ideal model.

Official scope references

Official scope: August 2026 Mechanical Engineers Licensure Examination Program — Professional Regulation Commission; Professional Regulatory Board of Mechanical Engineering — https://www.prc.gov.ph/sites/default/files/exam%20program%20aug%202026%20mele%20signed.pdf

Instructional references

Instructional reference: Introduction to Engineering Thermodynamics — Claire Yu Yan; BCcampus Open Education — https://pressbooks.bccampus.ca/thermo1/
Instructional reference: University Physics Volume 2 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-2
Instructional reference: How Hydropower Works — U.S. Department of Energy, Water Power Technologies Office — https://www.energy.gov/cmei/water/how-hydropower-works

BoardPrep lesson material is original instruction. It is not an official examination document.
No claim about question frequency or examination weighting is made.